

```
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/p8$ gcc -Wall -g memory_a
lloc.c -o memory_alloc
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/p8$ ./memory_alloc
=== Dynamic Memory Allocation ===

malloc(): allocated 5 integers
malloc values: 10 20 30 40 50

calloc(): allocated 5 integers
calloc values: 0 0 0 0 0

realloc(): resized array from 5 to 10 integers
After realloc: 10 20 30 40 50 60 70 80 90 100

free(): memory released successfully
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/p8$ |
```

```
free(): memory released successfully
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/p8$ valgrind --leak-check
=full ./memory_alloc
==2376== Memcheck, a memory error detector
==2376== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==2376== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==2376== Command: ./memory_alloc
==2376==
=== Dynamic Memory Allocation ===

malloc(): allocated 5 integers
malloc values: 10 20 30 40 50

calloc(): allocated 5 integers
calloc values: 0 0 0 0 0

realloc(): resized array from 5 to 10 integers
After realloc: 10 20 30 40 50 60 70 80 90 100

free(): memory released successfully
==2376==
==2376== HEAP SUMMARY:
==2376==    in use at exit: 0 bytes in 0 blocks
==2376==    total heap usage: 4 allocs, 4 frees, 1,104 bytes allocated
==2376==
==2376== All heap blocks were freed -- no leaks are possible
==2376==
==2376== For lists of detected and suppressed errors, rerun with: -s
==2376== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/p8$ |
```

```

tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/p8$ ./cow
Parent process PID: 3728
Allocated 1000000 integers.
Initial data[0] = 10

Forking process...

Child process PID: 3729
Child before modification: data[0] = 10
Child modifying data...
Child after modification: data[0] = 999
Child: Copy-on-Write page is created when memory is modified.

Parent process after child modification:
Parent data[0] = 10
Parent still has original value because of Copy-on-Write.
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/p8$ |

```

```

tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/p8$ ./cow
=====
          COPY-ON-WRITE DEMONSTRATION
=====
Parent PID: 5859
Allocated memory: 1000000 integers
Address of data: 0x72359fe2f010
Initial data[0]: 10

Forking process...

----- PARENT PROCESS -----
Parent PID: 5859
Child PID: 5860
----- CHILD PROCESS -----
Parent data before child modification: 10

Parent is waiting for child...
Child PID: 5860
Child before modification:
data[0] = 10
Address of data: 0x72359fe2f010

Child is modifying data[0]...

Child after modification:
data[0] = 999
Address of data: 0x72359fe2f010

Copy-on-Write occurred when the child modified the page.

Child will remain alive for 30 seconds.
Use another terminal to inspect /proc/5860/

```

```
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/p8$ cat /proc/5944/status
| grep -E "VmSize|VmRSS|VmData"
VmSize:      6680 kB
VmRSS:       4912 kB
VmData:      4132 kB
```

```
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/p8$ cat /proc/5944/maps
cat: /proc/5944/maps: No such file or directory
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/p8$ cat /proc/5984/maps
5c019801a000-5c019801b000 r--p 00000000 08:30 62397 /home/tarun/oslab/2520090020_Practical/p8/cow
5c019801b000-5c019801c000 r-xp 00001000 08:30 62397 /home/tarun/oslab/2520090020_Practical/p8/cow
5c019801c000-5c019801d000 r--p 00002000 08:30 62397 /home/tarun/oslab/2520090020_Practical/p8/cow
5c019801d000-5c019801e000 r--p 00002000 08:30 62397 /home/tarun/oslab/2520090020_Practical/p8/cow
5c019801e000-5c019801f000 rw-p 00003000 08:30 62397 /home/tarun/oslab/2520090020_Practical/p8/cow
5c01b0748000-5c01b0769000 rw-p 00000000 00:00 0 [heap]
75809cc2f000-75809d000000 rw-p 00000000 00:00 0
75809d000000-75809d020000 r--p 00000000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
75809d020000-75809d1c0000 r-xp 00002000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
75809d1c0000-75809d200000 r--p 001c0000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
75809d200000-75809d212000 r--p 0020d000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
75809d212000-75809d214000 rw-p 00211000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
75809d214000-75809d221000 rw-p 00000000 00:00 0
75809d3eb000-75809d3ee000 rw-p 00000000 00:00 0
75809d3f6000-75809d3f8000 rw-p 00000000 00:00 0
75809d3f8000-75809d3fc000 r--p 00000000 00:00 0 [vvar]
75809d3fc000-75809d3fe000 r--p 00000000 00:00 0 [vvar_vclock]
75809d3fe000-75809d400000 r-xp 00000000 00:00 0 [vdso]
75809d400000-75809d401000 r--p 00000000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
75809d401000-75809d430000 r-xp 00001000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
75809d430000-75809d43b000 r--p 00030000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
75809d43b000-75809d43d000 r--p 0003b000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
75809d43d000-75809d43e000 rw-p 0003d000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
75809d43e000-75809d43f000 rw-p 00000000 00:00 0
7ffcfddeb000-7ffcfded0000 rw-p 00000000 00:00 0
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/p8$ grep -E "heap|stack" /proc/5984/maps
5c019801a000-5c019801b000 r--p 00000000 08:30 62397 /home/tarun/oslab/2520090020_Practical/p8/cow
5c019801b000-5c019801c000 r-xp 00001000 08:30 62397 /home/tarun/oslab/2520090020_Practical/p8/cow
5c019801c000-5c019801d000 r--p 00002000 08:30 62397 /home/tarun/oslab/2520090020_Practical/p8/cow
5c019801d000-5c019801e000 r--p 00002000 08:30 62397 /home/tarun/oslab/2520090020_Practical/p8/cow
5c019801e000-5c019801f000 rw-p 00003000 08:30 62397 /home/tarun/oslab/2520090020_Practical/p8/cow
5c01b0748000-5c01b0769000 rw-p 00000000 00:00 0 [heap]
7ffcfddeb000-7ffcfded0000 rw-p 00000000 00:00 0 [stack]
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/p8$
```

```
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical/p8$ pmap 6048
6048:      ./cow
0000581969151000      4K r---- cow
0000581969152000      4K r-x-- cow
0000581969153000      4K r---- cow
0000581969154000      4K r---- cow
0000581969155000      4K rw--- cow
0000581991983000     132K rw--- [ anon ]
000076b2a422f000    3908K rw--- [ anon ]
000076b2a4600000     160K r---- libc.so.6
000076b2a4628000    1632K r-x-- libc.so.6
000076b2a47c0000     312K r---- libc.so.6
000076b2a480e000      16K r---- libc.so.6
000076b2a4812000       8K rw--- libc.so.6
000076b2a4814000      52K rw--- [ anon ]
000076b2a48b3000      12K rw--- [ anon ]
000076b2a48be000       8K rw--- [ anon ]
000076b2a48c0000     16K r---- [ anon ]
000076b2a48c4000       8K r---- [ anon ]
000076b2a48c6000       8K r-x-- [ anon ]
000076b2a48c8000       4K r---- ld-linux-x86-64.so.2
000076b2a48c9000    188K r-x-- ld-linux-x86-64.so.2
000076b2a48f8000     44K r---- ld-linux-x86-64.so.2
000076b2a4903000       8K r---- ld-linux-x86-64.so.2
000076b2a4905000       4K rw--- ld-linux-x86-64.so.2
000076b2a4906000       4K rw--- [ anon ]
00007ffcb3a54000    136K rw--- [ stack ]
total                6680K
```